

Bài tập 1:

Chúng ta có interface mail chứa các method sendMail(), log()
Class MailImpl sẽ implements mail và triển khai các method
mà method mà mail khai báo.

```
interface mail {  
    void sendMail();  
    void log();  
}  
  
class MailImpl implements mail {  
    @Override  
    public void sendMail() {  
        S.o.pln("Dang gui Mail...");  
    }  
    @Override  
    public void log() {  
        S.o.pln("Dang ghi log...");  
    }  
}  
  
class Main {  
    public static void main(String[] args) {  
        mail myMail = new MailImpl();  
        myMail.sendMail();  
        myMail.log();  
    }  
}
```


Bài tập 2:

```
Interface Shape {
    double get Area ();
    double get Perimeter ();
    void draw ();
}

class Rectangle implements Shape {
    @Override
    public double getArea (int width, int height) {
        System.out.println ("The area of the rectangle is " +
            (width * height));
    }
    @Override
    public double getPerimeter (int width, int height) {
        System.out.println ("The perimeter of the rectangle is
            " + (2 * (width + height)));
    }
    @Override
    public void draw () {
        System.out.println ("Drawing a Rectangle ...");
    }
    public boolean isSquare () {
        return width == height;
    }
}

class Circle implements Shape {
    @Override
    public double getArea () {
        return Math.PI * radius * radius;
    }
    @Override
    public double getPerimeter () {
        return 2 * Math.PI * radius;
    }
    @Override
    public void draw () {
        System.out.println ("Drawing a Circle ...");
    }
}
```



```

class Main {
    public static void main (String [] args) {
        Rectangle r1 = new Rectangle (5, 10);
        Circle c1 = new Circle (3.5);
        Shape [] shape = new Shape [2];
        shape [0] = r1;
        shape [1] = c1;
        System.out.println ("Diện tích : + s.getArea());
        System.out.println ("Chu vi : + s.getPerimeter());
    }
}

```

Bài tập 3:

```

interface Car {
    String getColor();
    double getSpeed();
    void accelerate();
}

class BMW implements Car {
    @Override
    public void accelerate() {
        this.speed += 20;
        System.out.println ("BMW");
    }
    @Override
    public String getColor() {
        return this.color;
    }
    @Override
    public double getSpeed() {
        return this.speed;
    }
    @Override
    public String toString() {
        return "BMW [màu: " + color + " tốc độ: " + speed + "]"
    }
}

```



```

class Civic implements Car {
    private String color;
    private double speed;
    public Civic (String color, double speed) {
        this.color = color;
        this.speed = speed;
    }
    @Override
    public void accelerate () {
        this.speed += 15;
        System.out.println ("Civic");
    }
    @Override
    public String getColor () {
        return this.color;
    }
    @Override
    public double getSpeed () {
        return this.speed;
    }
    @Override
    public String toString () {
        return "Civic [màu : " + color + " , tốc độ : " + speed
+ "]" ;
    }
}

```

```

class Main {
    public static void main (String [] args) {
        Car x1 = new Bmw ("Black", 100);
        Car x2 = new Civic ("White", 80);
        System.out.println ("Trước khi tăng tốc :");
        System.out.println (x1.toString());
        System.out.println (x2.toString());
        x1.accelerate();
        x2.accelerate();
        System.out.println ("Sau khi tăng tốc :");
        System.out.println (x1.toString());
        System.out.println (x2.toString());
    }
}

```


Bài tập 4:

```
public class HangSanXuat {  
    private String tenHangSanXuat;  
    private String tenQuocGia;  
    public HangSanXuat (String tenHangSanXuat, String  
tenQuocGia) {  
        this.tenHangSanXuat = tenHangSanXuat;  
        this.tenQuocGia = tenQuocGia;  
    }  
    public String getTenHangSanXuat () {  
        return tenHangSanXuat;  
    }  
    public String getTenQuocGia () {  
        return tenQuocGia;  
    }  
}  
public abstract class PhuongTienDiChuyen {  
    protected String loaiPhuongTien;  
    protected HangSanXuat hangSanXuat;  
    public PhuongTienDiChuyen (String loaiPhuongTien,  
hangSanXuat hangSanXuat) {  
        this.loaiPhuongTien = loaiPhuongTien;  
        this.hangSanXuat = hangSanXuat;  
    }  
    public String layTenHangSanXuat () {  
        return hangSanXuat.getTenHangSanXuat();  
    }  
    public void batDau () {  
        System.out.println ("Bat dau ...");  
    }  
    public void tangToc () {  
        System.out.println ("Tang toc ...");  
    }  
    public void dungLai () {  
        System.out.println ("Dung lai ...");  
    }  
    public abstract double layVanToc();  
}
```



```

class MayBay extends PhuongTienDiChuyen {
    private String loaiNhiemLieu;
    public MayBay (HangSanXuat hangSanXuat, String
    loaiNhiemLieu) {
        super ("May Bay", hangSanXuat);
        this.loaiNhiemLieu = loaiNhiemLieu;
    }
    public void eatCanh () {
        System.out.println ("Cat canh ...");
    }
    public void haCanh () {
        System.out.println ("Ha canh ...");
    }
    @Override
    public double layVanToc () {
        return 800.0;
    }
}

```

```

class XeOto extends PhuongTienDiChuyen {
    private String loaiNhiemLieu;
    public XeOto (HangSanXuat hangSanXuat, String
    loaiNhiemLieu) {
        super ("Xe Oto", hangSanXuat);
        this.loaiNhiemLieu = loaiNhiemLieu;
    }
    @Override
    public double layVanToc () {
        return 120.0;
    }
}

```

```

class XeDap extends PhuongTienDiChuyen {
    public XeDap (HangSanXuat hangSanXuat) {
        super ("Xe Dap", hangSanXuat);
    }
    @Override
    public double layVanToc () {
        return 20.0;
    }
}

```



```

}
public class Main {
    public static void main (String[] args) {
        HangSanXuat h1 = new HangSanXuat ("Boeing", "USA");
        HangSanXuat h2 = new HangSanXuat ("Toyota", "Japan");
        HangSanXuat h3 = new HangSanXuat ("Thong Nhat", "Vietnam");
        MayBay mb = new MayBay (h1, "Xang Jet A1");
        XeOto oto = new XeOto (h2, "Xang RON 95");
        XeDap xd = new XeDap (h3);
        System.out.println ("Phuong tien: " + mb.layTenHang
SanXuat ());
        mb.batDau ();
        mb.eatCanh ();
        System.out.println ("VanToc: " + mb.layVanToc ());
        System.out.println ("\n Phuong tien: " + oto.layTenHang
SanXuat ());
        oto.tangToc ();
        System.out.println ("VanToc: " + oto.layVanToc ());
        System.out.println ("\n Phuong tien: " + xd.layTen
HangSanXuat ());
        xd.dungLai ();
        System.out.println ("VanToc: " + xd.layVanToc ());
    }
}

```

Bài tập 5:

```

interface Moveable {
    void moveUp ();
    void moveDown ();
    void moveLeft ();
    void moveRight ();
}

class MoveablePoint (int x, int y, int xSpeed, int ySpeed) {
    this.x = x;
    this.y = y;
    this.xSpeed = xSpeed;
    this.ySpeed = ySpeed;
}

```



```
@Override  
public void moveUp() {  
    y -= ySpeed;  
}
```

```
@Override  
public void moveDown() {  
    y += ySpeed;  
}
```

```
@Override  
public void moveLeft() {  
    x -= xSpeed;  
}
```

```
@Override  
public void moveRight() {  
    x += xSpeed;  
}
```

```
@Override  
public String toString() {  
    return "Point at (" + x + ", " + y + ")";  
}
```

```
class MoveableCircle implements Moveable {  
    private int radius;  
    private MoveablePoint center;  
    public MoveableCircle(int x, int y, int xSpeed, int ySpeed,  
int radius) {  
        this.center = new MoveablePoint(x, y, xSpeed, ySpeed);  
        this.radius = radius;  
    }
```

```
@Override  
public void moveUp() {  
    center.moveUp();  
}
```

```
@Override  
public void moveDown() {  
    center.moveDown();  
}
```



```

@Override
public void moveLeft () {
    center.moveLeft ();
}

@Override
public void moveRight () {
    center.moveRight ();
}

@Override
public String toString () {
    return "Circle with radius " + radius + " centered  
at " + center.toString ();
}

class Main {
    public static void main (String[] args) {
        Moveable p1 = new MoveablePoint (5, 5, 2, 3);
        p1.moveUp ();
        p1.moveRight ();
        System.out.println ("Sau khi di chuyen: " + p1.toString ());
        Moveable c1 = new MoveableCircle (10, 10, 5, 5, 15);
        System.out.println ("\nVong tron ban dau: " + c1.  
toString ());
        System.out.println ("Vong tron sau khi di chuyen: " +  
c1.toString ());
    }
}

```